Question 1 Solution:
Given Information
RTT (user ↔ local DNS) = 5 ms
RTT (local DNS ↔ any other DNS server i.e., root/TLD/auth) = 30 ms
RTT (user ↔ web server www.company.com) = 150 ms
RTT (company email server ↔ local DNS) = 5 ms
RTT (company email server ↔ recipient's email server) = 100 ms
Web page: 1 base HTML + 10 images
Browser: no cache unless stated otherwise.

(I) Time required for DNS resolution

When local DNS does not have the answer, the local DNS will typically query root → TLD → authoritative sequentially (each is one RTT between local DNS and the other DNS server).
Time for the three external queries (local DNS ↔ any other DNS server) = 3 * 30 = 90 ms
Round trip time from user's host to local DNS: 5 ms
Total DNS time = 90 + 5 = 95 ms

(II) Web page loading time (no caching; HTTP/1.1 persistent + pipelining)

Sequence (serial, because the browser must get and parse base HTML to see image URLs):
Time for DNS resolution = **95 ms**
TCP connection establishment time to web server = 1 RTT = 150 ms
Request and receive base HTML (HTTP request → response) = 1 RTT= 150 ms
Note: Only after the base HTML arrives, the browser can learn the 10 image URLs
Browser issues pipelined GETs for the 10 images on the already-open connection; server sends back the image responses.
With pipelining the round-trip to request + receive all images is effectively = 1 RTT = 150 ms
Total Time (from click until all objects received)= **95 + 150 + 150 +1 150 = 545 ms**

Note: images cannot be requested until base HTML is parsed, so pipelining helps after the base HTML but does not remove the need to fetch the base HTML first.

(III) Effect of caching (local DNS has IP cached; browser has base HTML cached but not images)

Local DNS already has IP cached
(a) When there is no open TCP connection to the server

DNS time: one client↔local DNS query/response = **5 ms**
Need to fetch images (browser has base HTML locally, so no HTML fetch). Must open (or reuse) a TCP connection to the web server:
TCP handshake: **150 ms** (assume no existing TCP connection)
After handshake, browser issues pipelined GETs for 10 images = 1 RTT = **150 ms**
Total time = DNS time + TCP handshake + pipelined images = 5 + 150 + 150 = **305 ms**
(b) with an open persistent TCP connection to the server
The browser already had an open persistent TCP connection to the server, then there would be saving of 150 ms TCP handshake and 5 ms DND resolution
So, the total time would drop to **150 ms** (305 – 5- 150)

(IV) DNS resolution to get MX for example.net (local DNS has no cache)

local DNS → root: 1 RTT = 30 ms
local DNS → TLD: 1 RTT = 30 ms
local DNS → authoritative: 1 RTT = 30 ms
subtotal for external DNS = 3 * 30 = 90 ms
Finally, the company email server must query its local DNS and receive the answer:
Host↔local DNS RTT = **5 ms**
Total DNS resolution time = 90 + 5 = **95 ms**

(V) SMTP transaction time after DNS resolution

Sequence: the server first establishes a TCP connection, then the SMTP commands: EHLO, MAIL FROM, RCPT TO, DATA, and QUIT. Each command/response pair takes one RTT between the mail servers. TCP handshake is counted as 1 RTT.
RTT between mail servers = **100 ms**. (given)
RTTs counting
TCP handshake: 1 RTT = 100 ms
HELO → 250 reply: 1 RTT = 100 ms
MAIL FROM → 250 reply: 1 RTT = 100 ms
RCPT TO → 250 reply: 1 RTT = 100 ms
DATA → 354 (then send body) → 250: 1 RTT = 100 ms (DATA content is negligible but the command/response consumes an RTT)
QUIT → 221 reply: 1 RTT = 100 ms
Total RTTs = 6
Total SMTP transaction time = **6 * 100 = 600** ms for SMTP transaction after DNS resolution

Question 2 Solution:
(A)
Hosts (end systems): The houses and offices where letters are sent from and received.
Communication links: The roads, railways, or air routes that carry mail trucks, trains, or planes.

(B)

Layer	Output
Application Layer	Message
Transport Layer	Segment
Network Layer	Datagram
Data Link Layer	Frame

Question 3 Solution:
There is over-counting of time (1/b factor).
After time 1/b, the first bit will be on the wire.
This bit will reach the destination in time (d – (1/b))
Receiver can sense this bit in 1/b second
Receiver can sense the remaining (x – 1) bits in (x-1)/b seconds
Total time = 1/b + d – (1/b) + 1/b + (x-1)/b = d + x/b

Alternate explanation:
The leading edge of the physical wave form of the first bit will reach the destination after d seconds.
The trailing edge of the last bit signal will arrive at the destination after additional x/b seconds

Another solution (explanation): An extra 1/b time is counted for the following: Host B needs another 1/b1/b1/b second to sense the bit. In fact, the receiver does **not** need a full extra bit-time beyond propagation to sense the arriving bit. When the bit arrives at B, it is already on the link and can be observed/received. In the pipeline model the arrival of the first bit is at 1/b +d and the rest of the bits arrive continuously over the next (x−1)/b seconds, so the final arrival time is
(1/b + d) + (x-1)/b = d + x/b (with no extra +1/b)

Q4 Solution:

(A)

I. Cache hit ratio

Total requests / hour = 3600 + 2400 + 1200 = 7200

Hits per hour = requests for A + requests for B = 3600 + 2400 = 6000 / hour

Hit ratio = hits per hour / total requests per hour = 6000 ÷ 7200 = 5 / 6 = 0.83333 = **83.33%**

II. Bytes saved on the uplink per hour (because A and B are cached)

Bytes saved per hour = (size of A * requests for A per hour) + (size of B * requests for B per hour)

= (100000 bytes * 3600 requests)+(500000 bytes * 2400 requests)=1560000000 bytes/hour = **1560 MB / hour.**

(B)

Timeline	Scenario 1 (without interleaving)	Scenario 2 (with interleaving)
1st frame of Object A transmission complete at t =	4 ms (0.004 sec)	4 ms (0.004 sec)
Entire Object A transmission complete at t =	8 ms (0.008 sec)	12 ms (0.012 sec)
1st frame of Object B transmission complete at t =	12 ms (0.012 sec)	8 ms (0.008 sec)
Entire Object B transmission complete at t =	20 ms (0.02 sec)	20 ms (0.02 sec)

Q 5 Solution:

Packet size: L = 5,000 bits
No. of packets = 10,000/5,000 = 2
No. links = 2
Transmission rate of each link: $R = 10^6$ bps
Transmission delay for one packet = $d_{trans} = 5{,}000/10^6 = 5$ms
Processing delay at the switch: d_{proc} = 35 microseconds = 0.035ms
Propagation delay of each link: d_{prop} = 20 microseconds = 0.02ms

Time required to receive the 1st packet arrives at the destination = $d_{trans} + d_{prop} + d_{proc} + d_{trans} + d_{prop}$
=5ms+0.02ms+0.035ms+5ms+0.02ms
= 10.075ms

After transmitting the 1st packet, the source starts transmitting the 2nd packet. Since the switch implements store and forward scheme, 2nd packet arrives at destination at time = d_{trans}+10.075ms
=5ms+10.075ms
=15.075ms

Thus, it takes 15.075ms to receive 10,000 bits.